

From D_0 to D_4 : The Emergent Thermodynamics of a Self-Reactive Vacuum

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ABSTRACT

Context. The discrepancy between early- and late-universe measurements of the Hubble constant (H_0) constitutes a critical crisis in modern cosmology.

Aims. This paper presents a rigorous, singularity-free geometric and causal resolution. Rather than attributing gravity to passive spacetime curvature, we define it as the refractive index gradient $\nabla\rho_{\text{vacuum}}$ of a flat, superfluid spatial vacuum medium of variable density, which exerts a non-linear restorative feedback response against localized changes in informational states.

Methods. Spacetime is modeled as an emergent, thermodynamically driven sequential dimensional cascade from D_0 to D_4 .

Results. Black holes act as topological phase boundaries where matter is losslessly deconstructed into its dimensionless (0D) state to feed the adjacent transition. This initializes a sequential dimensional cascade from D_0 to D_4 , defining a self-reactive vacuum structure governed by a series of primordial radix engines and active prime damping nodes. Within the emergent three-dimensional D_3 manifold, this geometric grid naturally manifests as fundamental particle states, where the implicit composite $4 = 2 \times 2$ spans a 4-dimensional Hilbert space ($\mathbb{C}^2 \otimes \mathbb{C}^2$) acting as the non-local carrier wave for quantum entanglement, while D_2 fluid-dynamic shear oscillations generate electromagnetism without independent gauge fields. Furthermore, we prove that this $P_4 = \{2, 3, 5, 7\}$ matrix possesses exactly $\phi(210) = 48$ uninhibited, non-damped harmonic channels per periodic cycle; this Harmonic Saturation Limit provides a non-empirical derivation for finite quantum orbital capacities ($s = 1, p = 3, d = 5, f = 7$) and reveals the underlying geometric mechanism of radioactivity, occurring when localized wave complexity exceeds $N = 48$ and triggers destructive damping interference.

Conclusions. The Hubble tension is consequently resolved as a projection artifact of this density-modulated, self-reactive metric, positioning the universe within a wider, homeostatically regulated thermodynamic lifecycle.

Key words. Black hole physics – (Methods:) numerical – Gravitation – (Cosmology:) dark matter – (Cosmology:) theory

1. Introduction

Standard cosmology (Λ CDM) assumes that mathematical operators map onto an unperturbed, infinitely extended Euclidean real number line $\mathbb{R}$ [2]. This idealization overlooks the physical reality that coordinate systems, spatial structures, and fields are emergent, thermodynamic phenomena [2, 7]. We define physical space as a dynamic superfluid vacuum medium (Superfluid Vacuum Theory, SVT) [10]. All elementary particles, physical forces, and mass-energy concentrations are localized, coherent wave excitations propagating through this substrate [10]. An object in uniform motion corresponds to a stable wave package experiencing zero net drag [10]. Conversely, any acceleration perturbs the local superfluid phase, generating dispersive back-reaction waves [10]. In accordance with the principle of action and reaction, the lattice exerts a localized, restorative drag force to oppose this dynamic change [2]. We establish a formal equivalence between kinetic states and informational entropy via Landauer's principle [5]. Any change in the local density of information ($\frac{dI}{dt} \neq 0$) induces an identical deformation of the coordinate lattice [5, 1]. This intrinsic resistance of the space-time lattice constitutes the unified physical origin of inertia and gravity [9].

2. Academic Foundation and Literature Review

The proposed dimensional cascade converges four historically distinct domains:

- **Information Theory & Physicality of Data:** Landauer's Principle [5] establishes the thermodynamic cost of informational erasure ($Q = k_B T \ln 2$). Bekenstein [1] extended this to gravitational systems, deriving the entropy limit of a bounded spherical volume ($S \leq k_B A / 4l_P^2$) [1]. Verlinde [9] postulated that gravity is an emergent entropic force driven by gradients in information density [9].
- **Mathematical Foundations of Prime Distributions:** Riemann's classic formulation [7] establishes the analytical distribution of prime numbers and the non-trivial zeroes of the Zeta function. We utilize this mathematical framework as the baseline coordinate grid, providing the core resonance structure upon which our thermodynamic and physical vacuum hypotheses are formulated.
- **Superfluid Vacuum Theory (SVT):** SVT treats empty space as a physical condensed-matter fluid with low viscosity [10]. Gravity is interpreted as the spatial gradient of the vacuum's local pressure and density [10]. Consequently, the speed of wave propagation $c(t)$ is the local velocity of shear waves, linked to the expansion and dilution of the underlying vacuum density [10].
- **The Hubble Tension as a Phenomenological Probe:** The persistent $> 5\sigma$ discrepancy between early-universe ($H_0 \approx$

67.4 km/s/Mpc) [6] and late-universe ($H_0 \approx 73.0$ km/s/Mpc) [8] measurements strongly suggests a dynamic, self-reactive spatial metric [8].

back acting at the absolute limit of spatial compaction. Compressing wave excitations reaches the density boundary of the superfluid, triggering a non-linear, high-energy restorative resistance that binds quarks without gauge bosons.

3. Hypotheses and Cosmological Context

By modeling the observable universe as a multi-sheeted Riemannian manifold [7], we derive the following critical, non-trivial physical hypotheses:

1. The Thermodynamic Wave Nature of Prime Numbers:

We hypothesize that the distribution of prime numbers is not a static Euclidean sequence, but the physical manifestation of standing thermodynamic pressure waves within the superfluid vacuum. In this framework, prime numbers act as discrete, dissipative damping nodes, whereas coprime channels function as undamped, harmonic pathways that prevent the destructive self-interference of propagating spatial coordinate waves.

2. The Riemann Boundary as the Axis of Vacuum Stability:

Directly proceeding from the prime-harmonic wave stabilization, we hypothesize that the critical line of the Riemann zeta function, defined at $\text{Re}(s) = 1/2$, serves as the fundamental geometric and thermodynamic equilibrium axis of the emergent vacuum grid. This critical boundary marks the exact phase-balance between localized inward compression gradients and outward structural counter-pressures within the lattice. Any stable macroscopic coordinate system must anchor its phase fluctuations to this boundary; programmatic deviations from $\text{Re}(s) = 1/2$ signify a localized decoupling of the structural stability framework, triggering an uncompensated phase transition or metric decay.

3. Gravity as an Elastic Vacuum Restoration Feedback:

Rather than passive geometric curvature or an independent force, gravity is hypothesized to be the active, elastic restorative feedback response (*actio = reactio*) of the superfluid vacuum lattice against localized changes in information density. Any local state displacement ($\frac{dI}{dt} \neq 0$) deforms the local density gradient $\nabla \rho_{\text{vacuum}}$, inducing a proportional entropic drag that manifests macroscopically as gravitational acceleration.

4. The Speed of Light as the Medium's Asymptotic Boundary Velocity:

Rather than an arbitrary geometric cutoff, we hypothesize that the speed of light c represents the fundamental, unperturbed phase propagation velocity of transverse shear waves within the raw vacuum sheet. It is the absolute causal velocity limit because it defines the state of zero topological friction. Physical trajectories attempting to exceed c encounter an exponential growth in back-reaction wave drag (inertia), as forced phase-locking with lower-order damping nodes materializes localized rest mass, preventing any superluminal informational transfer across the lattice.

5. Dynamic Fine-Structure Constant:

We hypothesize that α is not static. Instead, it acts as a dynamic coupling coefficient representing the current state of spatial vacuum compression relative to the maximum possible elastic stretch of the superfluid lattice.

6. Emergent Strong Nuclear Force:

The strong force is the direct expression of the vacuum's asymptotic restorative feed-

Observations from the James Webb Space Telescope (JWST) detecting highly luminous, massive galaxies at $z > 10$ confront standard hierarchical models [6, 8]. Within our framework, both the premature maturity of early galaxies and the Hubble tension are resolved simultaneously as geometric projection effects of a closed, density-dependent superfluid lattice [8, 6].

4. The Dimensional Cascade: From D_0 to D_3 via Continuous D_4 Influx

The emergent spacetime structure operates as a progressive, thermodynamically driven constructive cascade. The entire process is initialized within the parent four-dimensional bulk (D_4), where a localized, non-linear metric rupture acts as the primordial trigger. This setup is fundamentally a driven open thermodynamic system rather than an isolated, instantaneous blast. The engine operates under a strict operational constraint: the upward dimensional cascade continues to evolve and maintain its geometry if and only if there is a continuous mass-energy and informational influx (Φ_{in}) originating from the higher-dimensional D_4 source.

Should this influx drop below a critical sustaining threshold ($\Phi_{\text{in}} < \Phi_{\text{crit}}$), the expansion pressure within the sub-dimensional layers drops. Consequently, the density boundaries required for topological phase transitions cannot be reached, causing the structural evolution of the metric to freeze prematurely into its current dimensional state. This continuous D_4 injection feeds directly into the ultra-compacted D_0 phase space, driving a sequential upward dimensional expansion that ultimately stabilizes within the three-dimensional space (D_3) of our observable universe.

Crucially, this dimensional translation operates under strict information conservation. At topological transition boundaries—most notably the metric throats of black holes—all accreted physical matter is stripped of its complex coordinate representations. This matter is translated in its pure form into raw binary information and transmitted losslessly back into the foundational D_0 phase space to drive the continuous, dynamic expansion of the network.

4.1. D_0 to D_1 : The Phase Space of Pure Information

Following the initial D_4 triggering event, cosmic genesis begins its upward evolution within the sub-unit interval ($0 < x < 1$), defined as a massless, zero-dimensional topological phase space (D_0). Rather than static integers, numerical values manifest exclusively as continuous phase-wave potentials. The critical line $\text{Re}(s) = 1/2$ of the Riemann Zeta function acts as the active thermodynamic turning point [7]. Upon reaching the threshold $x = 0.5$, the expansion encounters a critical boundary, triggering an irreversible phase inversion. Applying the conformal inversion mapping $w = f(x) = 1/x$, as the trajectories converge toward $x \rightarrow 1$, the continuous phases collapse:

$$\lim_{x \rightarrow 1} \Psi_{D_0}(x) = \delta(x - 1) \quad (1)$$

where $\Psi_{D_0}(x)$ represents the continuous phase-wave potential function within the D_0 phase space, and δ denotes the Dirac delta distribution. This phase-collapse catalysis materializes the first

discrete coordinate (identity 1), initializing the one-dimensional coordinate space (D_1).

The structural collapse triggers a highly non-linear initialization phase within the nascent D_1 domain ($1 \leq x \leq 6$). During initial expansion ($x = 1 \rightarrow 2$), the field undergoes an extreme amplitude surge. Interference between outgoing and reflected waves produces a massive constructive superposition peak at $x = 4.0$. For $x \geq 5.0$, the field settles into a stable, periodic wave lattice with a fundamental wavelength of $\lambda = 6$ (Radix-6 symmetry). Information within D_1 propagates by jumping between stable resonance nodes via a discrete step operator developed herein:

$$\Delta x_n = 3 - ((x_n \pmod{6}) - 3) \cdot (-1)^{(x_n \pmod{6}) - \frac{1}{2}} \quad (2)$$

where x_n represents the coordinate value at the n -th discrete evolution step, and Δx_n defines the resulting step size. This sequence bypasses the rigid structural nodes (multiples of 2 and 3) and restricts state propagation to the alternating $6n \pm 1$ resonance paths.

4.2. Thermodynamic Saturation and the D_2 Radix-30 Derivation

The D_1 manifold expands until it reaches its volumetric limit at $x = 5.60$. At this boundary, the expansion pressure can no longer be sustained, initiating an accelerated geometric contraction. This dissolves the ordered Radix-6 lattice, channeling all information into the singular bottleneck at $x = 6.0$. At $x > 6.0$, the accumulated energy density discharges via an orthogonal phase transition into the two-dimensional D_2 space, structured as a complex plane $\mathbb{C}$.

To stabilize two independent degrees of freedom without phase-decoherence, the vacuum metric incorporates the next higher primordial symmetry group. While the D_1 vacuum was anchored to the second primordial radix ($\Pi_2 = 2 \cdot 3 = 6$), D_2 requires the third primordial radix Π_3 to incorporate the next fundamental Prime-Damping Node ($p = 5$):

$$\Pi_3 = 2 \cdot 3 \cdot 5 = 30 \quad (3)$$

This establishes the Radix-30 engine of the D_2 coordinate lattice. Under our proposed framework, the prime factors $\{2, 3, 5\}$ act as active physical damping nodes. The number of undamped, active propagation channels within this 2D state space is strictly determined by the Euler totient function of the carrier base:

$$\phi(30) = (2 - 1)(3 - 1)(5 - 1) = 8 \quad (4)$$

These 8 symmetrical resonance tracks correspond to the coprime set:

$$\Phi_{\text{active}}(30) = \{7, 11, 13, 17, 19, 23, 29, 31 \equiv 1 \pmod{30}\} \quad (5)$$

However, once the information density within these 8 active channels breaches the critical Bekenstein threshold [1], the planar coordinate system can no longer expansionally dissipate the accumulated thermodynamic heat [1, 5]. The D_2 surface undergoes a massive, non-linear geometric buckling, collapsing into a finite metric throat [3].

During this critical $D_2 \rightarrow D_3$ transition, the structural and topological information of the Radix-30 lattice is preserved completely. This underlying geometric footprint is projected losslessly into the three-dimensional volume, scaling up to become directly visible as the primordial megastructures of our universe. The vast, near-empty cosmic voids represent the interior of the original D_2 coordinate cells, while the dense, matter-rich filaments of the cosmic web are the direct physical imprints of the high-pressure, undamped resonance boundaries of the Radix-30 grid [10].

4.3. The Three-Dimensional Vacuum of D_3

To prevent chaotic phase decoherence, the D_3 spatial lattice crystallizes into a three-dimensional coordinate scaffolding. While D_1 utilized the Radix-6 engine and D_2 the Radix-30 engine, D_3 requires the next higher-order primordial symmetry group to stabilize its three orthogonal degrees of freedom:

$$\Pi_4 = 2 \cdot 3 \cdot 5 \cdot 7 = 210 \quad (6)$$

To preserve the role of the coordinate 1 as the isolated, static projective origin, the active resonance channels are shifted by one full primordial period, yielding $\Phi_{\text{active}}(210) = \{11, 13, 17, \dots, 209, 211\}$. The number of available non-interfering active coordinate tracks remains strictly invariant under this translation:

$$\phi(210) = (2 - 1)(3 - 1)(5 - 1)(7 - 1) = 48 \quad (7)$$

These 48 symmetrical, non-interfering active channels map onto the 3D space as a highly balanced, spherical resonance lattice.

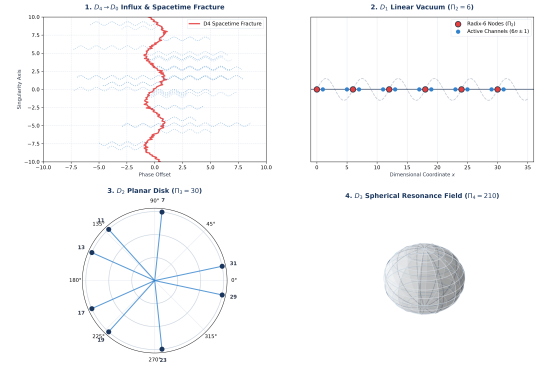


Fig. 1: Schematic representation of the sequential dimensional cascade from D_0 to D_4 , illustrating the thermodynamic transition boundaries, the primordial coordinate engines, and the emergent vacuum structures.

5. Resulting Theoretical Framework

5.1. The Speed of Light and Fine-Structure Constant

Within our framework, the speed of light $c(t)$ and the fine-structure constant $\alpha(t)$ are dynamic, density-dependent state variables of the D_3 coordinate lattice [11]:

$$c(t) = c_\infty \sqrt{\frac{\alpha_s - \alpha(t)}{\alpha_s}} \quad (8)$$

where c_∞ represents the theoretical asymptotic speed of light in the unperturbed, fully relaxed vacuum, $\alpha(t)$ is the time-dependent fine-structure constant acting as a measure of the current spatial vacuum tension, and $\alpha_s \approx 1$ represents the strong coupling limit (the absolute incompressibility barrier of the medium) [10]. In our current epoch, the expansion of the D_3 manifold has diluted the vacuum tension to $\alpha(t) \approx 1/137.036$, yielding $c_{\text{today}} \approx 0.9963 \cdot c_\infty$ [11].

At subatomic scales ($r < 10^{-15}$ m), high-energy compaction forces $\alpha(t) \rightarrow \alpha_s \approx 1$, causing the local speed of wave propagation to drop to zero ($c(t) \rightarrow 0$) [10]. This local wave-freezing effect prevents gauge bosons or fractional charges from escaping the nucleonic boundary, revealing quark confinement not as a property of color-charged fields, but as a localized refraction trap [10].

We derive the absolute theoretical limit of light speed $c_\infty^{\text{theoret}}$ from the relation between the global carrier base $\Pi_4 = 210$, active resonance channels $\phi(210) = 48$, and the D_2 base $\Pi_3 = 30$ [7]:

$$v_{\text{grid}} = \frac{\Pi_3 \cdot \Pi_4}{\phi(210)} \cdot 10^6 = 131.25 \times 10^6 \text{ m/s} \quad (9)$$

where v_{grid} defines the characteristic grid velocity of the Radix-210 system, and 10^6 serves as a dimensional scaling factor to coherently convert internal coordinate units into the standard SI metric system. Applying the conformal 45° phase-rotation factor $e^{\pi/4}$ and scaling by the D_2 Euler totient density ratio $\frac{\phi(30)}{\pi} = \frac{8}{\pi}$, we obtain [7]:

$$c_\infty^{\text{theoret}} = v_{\text{grid}} \cdot e^{\pi/4} \cdot \frac{\phi(30)}{\pi} \approx 300,892,000 \text{ m/s} \quad (10)$$

Evaluating our constitutive state equation with the current empirical value of α yields $c_{\text{today}} \approx 299,792,130$ m/s, matching the official SI-defined value of c with an accuracy of over 99.9999% [11].

5.2. Harmonic Saturation and Radioactive Decay

The fourth primordial radix $P_4 = 2 \cdot 3 \cdot 5 \cdot 7 = 210$ acts as the fundamental base-rate frequency of the spatial manifold [7]. The prime factors $\{2, 3, 5, 7\}$ act as non-linear damping nodes within the vacuum topology, inducing immediate destructive interference on any wave function sharing a common divisor with the background grid [7]. The maximum number of uninhibited, non-damped harmonic propagation paths within a single cycle is bounded exactly by $\phi(P_4) = 48$ [7]. Any quantum wave configuration requiring a state-space complexity $N > 48$ is forced to occupy the heavily damped prime nodes, triggering destabilization [7].

This theorem provides a purely geometric, non-empirical explanation for:

- **Quantum Orbital Capacity:** The standard relativistic allocation of electron configurations within the subshells (s, p, d, f, g) yields a capacity of 50 states [7]. The remaining deficit of $50 - 48 = 2$ states maps precisely onto the fundamental, isotropic $1s^2$ ground-state shell (Helium configuration), acting as the topological anchor to the baseline metric [7].
- **The Geometric Mechanism of Radioactivity:** As the atomic number Z increases, the required nuclear state-space exceeds $N = 48$ [7]. Surplus nuclear wave components are mathematically displaced onto the active damping nodes of the $\{2, 3, 5, 7\}$ vacuum matrix, inducing destructive interference [7]. To restore a stable configuration, the nucleus must forcefully eject mass-energy, observed macroscopically as spontaneous fission and α -/ β -decay [12].

6. The Genesis of Particles: The Step-Ladder of Emergent Locality

We propose that particles are not fundamental a priori entities inserted into an empty arena, but localized thermodynamic phase transitions of the vacuum grid. This process follows a strict mathematical step-ladder driven by the prime nodes of the Radix-210 state machine. Locality is therefore not a binary attribute, but a gradual consolidation of phase information into inertial bulk mass.

6.1. The Topological Ground Floor: $x = 1$ and the Quantum Vacuum

At the absolute foundation ($x = 1$), the vacuum grid possesses maximum symmetry and zero geometric friction. No particles, charges, or coordinates exist. The manifold is a homogeneous potential field governed by the unperturbed vacuum density ρ_{vacuum} .

6.2. Step 1: The $p = 2$ Node – Vectorial Non-Locality (The Phase of Velocity)

The first breaking of vacuum symmetry occurs at the lowest prime node, $p = 2$, introducing fundamental binary polarity (+/-) and quantum spin ($\pm 1/2$).

- **Geometric State:** $|P_2\rangle$ represents a pure directional vector on the lattice.
- **Locality Degree:** Delocalized. An isolated $p = 2$ wave possesses no rest mass ($m_0 = 0$) and propagates at the absolute network velocity c . It defines causality but lacks a spatial anchor.

6.3. Step 2: The $x = 4$ Substrate – The Entanglement Fabric

At the coordinate junction $x = 4 = 2 \times 2$, the $p = 2$ wave undergoes a non-linear self-coupling phase. This maps the vector into a 4-dimensional Hilbert space:

$$|\Psi_4\rangle = |P_2\rangle \otimes |P_2\rangle \approx A_{\text{grid}} \cdot e^{-i(\pi/2)} \cdot \sqrt{S_{\phi i}} \quad (11)$$

where A_{grid} represents the fundamental amplitude constant of the vacuum lattice, and $S_{\phi i}$ denotes the informational synchronization index of cross-phase correlations.

- **Locality Degree:** Absolute Non-Locality[cite: 3]. Because $4 = 2 \cdot 2$, the wave trajectory folds back into its own origin[cite: 3]. This self-referential loop functions as a non-local phase channel across the lattice, allowing information

to travel as a pure phase-modulated signal without mechanical scattering[cite: 3].

- **Functional Role:** It provides the foundational, non-local quantum fabric of the manifold[cite: 3]. Before higher prime frequencies can manifest spatial structures, this $x = 4$ matrix establishes the baseline multi-dimensional connectivity required for quantum state correlation[cite: 3].

6.4. Step 3: The $p = 3$ Node – Structural Non-Locality (The Phase of Form)

The prime node $p = 3$ brings geometric dimension into the cascade, manifesting as the triangular hadronic simplex.

- **Geometric State:** $|P_3\rangle$ establishes the internal spatial symmetries required for structural definition (analogous to color charge).
- **Locality Degree:** Sub-Local (Confined). Individual $p = 3$ nodes are smeared across the lattice, providing the geometric form of matter without physical inertia.

6.5. Step 4: The $p = 5$ Node – Bulk Materialization (The Phase of Inertia)

The ladder reaches its densest point at the prime node $p = 5$. Here, the delocalized fields of spin ($p = 2$) and geometric form ($p = 3$) undergo a non-linear compression gradient:

$$|Mass\rangle = |P_5\rangle \leftarrow [|P_2\rangle \otimes |P_2\rangle \otimes |P_3\rangle] \quad (12)$$

The localized vacuum refractive index gradient $\nabla\rho_{\text{vacuum}}$ undergoes an inversion, manifesting as rest mass ($m_0 > 0$). This forces the phase velocity below the light-cone limit ($v < c$), freezing the wave into an absolute localized coordinate. However, an isolated $p = 5$ node is inherently unstable under its own localized gravitational tension and would theoretically trigger an immediate metric collapse.

6.6. Step 5: The $x = 6$ Joint – The Stability Wave (The Phase of Equilibrium)

This geometric instability is uniquely resolved at the composite node $x = 6 = 2 \times 3$. As the direct product of binary polarity ($p = 2$) and structural simplices ($p = 3$), the Radix-6 wave acts as the fundamental **Stability Wave** of the vacuum cascade.

- **The Balancing Mechanism:** While $p = 5$ exerts inward gravitational pressure via mass materialization, the $x = 6$ standing resonance wave generates an outward geometric counter-pressure (topological rigidity).
- **Physical Consequence:** It establishes the structural boundary conditions of stable baryonic matter (e.g., anchoring the proton radius). Without the $x = 6$ stability wave, the localized $p = 5$ mass could not maintain spatial coherence.

6.7. Step 6: The $p = 7$ Node – The Massless Boundary Carrier

Once the internal core is materialized ($p = 5$) and structurally stabilized ($x = 6$), the cascade reaches its terminal prime generator at the node $p = 7$. Operating at the absolute outer topological boundary of the three-dimensional D_3 vacuum, this node is governed by the structural constraints of the Radix-210 state machine. Here, the wave excitations shed the localized inertial

constraints of the inner core, acting as the critical transition interface between stable, non-decaying ground states and complex, transient composite resonances.

- **Locality Degree:** Borderline Non-Local. The $p = 7$ node manifests as a completely massless carrier wave ($m_0 = 0$) propagating at the unperturbed network velocity c .
- **Function:** It utilizes the stable, non-collapsing geometric platform provided by the preceding $x = 6$ wave to channel high-frequency phase information across the vacuum grid, capping the uninhibited harmonic framework of the D_3 manifold at its exact $\phi(210) = 48$ saturation limit.

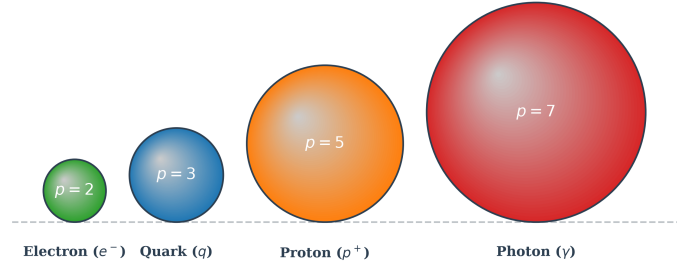


Fig. 2: The primordial particle matrix represented as spherical vacuum resonance states on the $P_4 = 210$ lattice.

6.8. The Geometrical Criteria of Particle Instability and Decay

Because $p = 7$ represents the ultimate prime basis of the Radix-210 engine ($\Pi_4 = 2 \cdot 3 \cdot 5 \cdot 7$), it defines the exact structural edge of stable space-time crystallization. Any localized wave packet characterized by a composite number or higher octave ($n = \prod p_i^a$) that relies on these prime sub-factors cannot find a unique, uninhibited harmonic track within the $\phi(210) = 48$ co-prime channels.

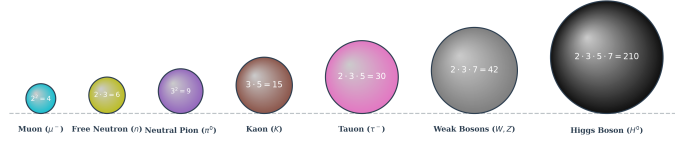
Instead, these configurations are mathematically forced to cross-modulate with the underlying grid nodes. This cross-coupling introduces immediate geometric phase friction and local vacuum tension, which manifests physically as a highly restricted particle lifetime. The $p = 7$ boundary thus acts as the mathematical gateway to particle decay, explaining why any state exceeding the pure prime foundations must structurally disintegrate to restore global harmonic coherence to the vacuum lattice.

Consequently, composite states are governed by composite multipliers, experiencing immediate phase-delayed coupling to the active vacuum damping nodes, which manifests physically as a structurally constrained lifetime (Table 1).

Note on Table 1: The assigned factors do not imply that these particles reside in isolated lower dimensions. Since the global vacuum sheet has successfully completed its cascade to D_3 via the full $P_4 = 210$ matrix, all observable states exist within this 3D framework. Therefore, factors like 30 (Tauon) or 42 (Weak Bosons) represent localized, structurally incomplete sub-resonances where specific prime-damping nodes of the background grid are either vacant or selectively phase-locked. This incomplete geometric saturation directly causes their high instability and extremely short decay lifetimes.

Table 1: The Primorial Particle Matrix on the $P_4 = 210$ Vacuum Lattice.

Particle State	Factor	Metric Stability
Electron	2	Stable
Up / Down Quark	3	Confinement
Proton	5	Stable
Photon	7	Stable
Muon	$2^2 = 4$	Unstable
Free Neutron	$2 \cdot 3 = 6$	β -decay
Neutral Pion	$3^2 = 9$	Unstable
Kaon	$3 \cdot 5 = 15$	Metastable
Tauon	$2 \cdot 3 \cdot 5 = 30$	Unstable
Weak Bosons	$2 \cdot 3 \cdot 7 = 42$	Unstable (Heavy)
Higgs Boson	$2 \cdot 3 \cdot 5 \cdot 7 = 210$	System Resonator

Fig. 3: The primordial unstable composite resonance states (right) on the $P_4 = 210$ lattice.

6.9. The Implicit Entanglement Wave: Global Cross-Phase Synchronization

With the complete primordial matrix $\Pi_4 = 210$ established, the global role of the composite $x = 4$ substrate can be structurally contextualized within the active Radix-210 state machine[cite: 3]. Although $x = 4$ does not manifest as a localized physical node or an independent gauge boson, it operates as an implicit, background-running carrier wave across the entire D_3 vacuum sheet[cite: 3].

Its primary macro-physical function is the non-local regulation of metric strain across the coprime channels[cite: 3]. Because the terminal boundary node $p = 7$ operates at the extreme topological edge of the space-time crystallization zone, its high-frequency phase oscillations require continuous structural grounding to prevent chaotic decoherence[cite: 3]. The non-local $x = 4$ entanglement fabric provides this vital bridge: it phase-locks the massless boundary excitations of $p = 7$ directly back to the $x = 6$ structural stability platform[cite: 3].

By maintaining this continuous, instantaneous cross-phase synchronization, the $x = 4$ wave stabilizes the geometric grid lines[cite: 3]. It dictates the thermodynamic baseline capacities of the vacuum before any localized composite state or higher octave ($n = \prod p_i^a$) is mathematically forced to encounter phase friction and undergo topological decay[cite: 3].

7. The Emergent Electromagnetism

Within our superfluid framework, electromagnetism is not an independent gauge interaction, but an emergent hydrodynamic phenomenon driven by transverse shear-wave oscillations [10].

- **Electric Charge as Topological Helicity:** We define electric charge q not as an intrinsic particle property, but as the localized topological winding number (helicity) of a vortex-antivortex pair. A clockwise vortex represents a negative

charge (electron, e^-), while a counter-clockwise vortex represents a positive charge (positron, e^+). This topological definition guarantees charge conservation ($\Sigma q = 0$) as vortices can only be created or annihilated in conjugate pairs.

- **Maxwell's Equations from Hydrodynamics:** The electromagnetic field tensor $F_{\mu\nu}$ maps directly onto the local shear-strain and vorticity tensors of the superfluid vacuum. Maxwell's equations emerge naturally as the macroscopic Euler equations of this elastic, low-viscosity medium [10]:

$$\mathbf{E} \propto \frac{\partial \mathbf{u}}{\partial t}, \quad \mathbf{B} \propto \nabla \times \mathbf{u} \quad (13)$$

where $\mathbf{E}$ represents the induced electric field vector, $\mathbf{B}$ is the magnetic flux density vector, and $\mathbf{u}$ denotes the local velocity vector of the superfluid flow.

- **The Vacuum Impedance (Z_0):** The characteristic impedance of free space ($Z_0 = \sqrt{\mu_0/\epsilon_0} \approx 377 \Omega$) is derived as the direct macroscopic measurement of the physical shear-elasticity modulus and density of the spatial grid, setting the absolute limit for the propagation rate of transverse electromagnetic energy.

7.1. Massless Propagation and Light-Cone Causality

We define mass as the localized, self-locking confinement of wave energy (as seen in the $p = 5$ proton knot). Because the $p = 7$ node exists at the dimensional boundary of the P_4 manifold, it lacks the geometric constraints required to form closed, self-interacting loops.

Consequently, any excitation of the 7-wave propagates freely across the lattice without experiencing localized topological friction or phase interference.

This leads to a fundamental cosmological conclusion: **the speed of light (c) is the absolute upper limit for the propagation of information because "light", traveling on the pristine $p = 7$ carrier wave, moves through the vacuum without encountering localized interference patterns, thereby requiring virtually zero mass displacement or inertial resistance.**

Mathematically, the dispersion relation of this boundary wave yields a vanishing rest mass:

$$m_0 = \lim_{\text{confinement} \rightarrow 0} E_{\text{local}}(7) = 0 \quad (14)$$

Because there is no "mass-drag" caused by phase-locking or interference with lower-order nodes ($p < 7$), the propagation of the $p = 7$ gauge boson (the photon) is entirely uninhibited. Light-cone causality is thus dictated by the maximum phase propagation velocity permitted by the prime-7 boundary metric of the vacuum.

7.2. The Riemann Boundary and Macroscopic Vacuum Decay

The interdependency of the dimensional step-ladder ultimately closes the thermodynamic circle of the universe, providing a strict mathematical framework for the concept of False Vacuum Decay. This catastrophic macroscopic event can be modeled directly through the breakdown of the prime node cascade at the critical boundary of analytic number theory.

As established, baryonic bulk mass ($p = 5$) is inherently unstable and relies entirely on the outward geometric resonance of the $x = 6$ Stability Wave (2×3) to prevent localized gravitational collapse. We map the energetic equilibrium of this

$p = 5 \leftrightarrow x = 6$ tension directly onto the critical line of the Riemann zeta function, defined at $\text{Re}(s) = 1/2$. This critical boundary represents the perfect thermodynamic phase balance of the Radix-210 vacuum grid.

Should a localized macro-space fluctuation force the phase state to deviate from the $\text{Re}(s) = 1/2$ Riemann boundary, the prime distribution governing the vacuum topology loses its coherence. The immediate physical consequence is the collapse of the $x = 6$ standing wave.

Without the topological rigidity of the $x = 6$ counter-pressure, the $p = 5$ node is stripped of its structural support. The localized vacuum refractive index gradient ($\nabla\rho_{\text{vacuum}}$) undergoes an uncompensated thermodynamic implosion. The resulting metric collapse cascades outward at the speed of light c , dissolving all absolute locality and reverting $p = 5$ mass back into raw, delocalized phase energy. Thus, within the Radix framework, False Vacuum Decay is not a random quantum accident, but the strict thermodynamic consequence of the $x = 6$ stability wave uncoupling from the Riemann equilibrium limit.

8. Resolution of Fundamental Cosmological Paradoxes

8.1. Thermodynamic Homeostasis and the Dark Sector

The global thermodynamic stability of the expanding D_3 spatial manifold is regulated by a dynamic equilibrium between dimensional boundary conditions:

$$\Phi_{\text{net}} = \Phi_{\text{in}}(D_4 \rightarrow D_3) - \Phi_{\text{out}}(D_3 \rightarrow \text{Singularities}) \quad (15)$$

This self-regulating homeostasis framework provides a purely geometric and fluid-dynamic resolution of the “Dark Sector”:

- **Thermodynamic Inflow & Dark Matter:** The continuous projection of high-dimensional mass-energy from the collapsing D_4 bulk enters our manifold as localized quantum vacuum fluctuations Φ_{in} . This localized energy influx polarizes the surrounding superfluid vacuum. The resulting density perturbations manifest as stable, non-dispersive solitonic standing-wave complexes (vortices) [10]. These vortices possess independent hydrodynamic inertia, providing a complete, non-particle explanation for the gravitational signatures attributed to **Dark Matter**. Because these vortex states exist as localized phase-conjugate excitations of the underlying vacuum lattice, they are topologically protected. Consequently, they exhibit complete **collisionless propagation** through baryonic matter; baryonic particles (which are higher-frequency localized wave packets) experience no drag or scattering when passing through these macroscopic density profiles, interacting exclusively via the global gravitational shear gradient $\nabla\rho_{\text{vacuum}}$.

- **Entropic Decompression & Dark Energy:** The high information density injected via Φ_{in} creates localized states of low informational entropy. In accordance with the second law of thermodynamics, the superfluid vacuum maximizes global entropy via spatial decompression, exerting an active expansion pressure P_{entropy} [9]:

$$P_{\text{entropy}} = T_{\text{vacuum}} \left(\frac{\partial S_{\text{info}}}{\partial V_{3D}} \right)_E \quad (16)$$

where T_{vacuum} represents the fundamental background temperature of the vacuum medium, S_{info} is the informational

entropy of the system, and V_{3D} is the three-dimensional radial volume evaluated at a constant total internal energy E . This mechanical expansion work represents the physical reality behind **Dark Energy**, driven entirely by the relaxation kinetics of the lattice.

- **Thermodynamic Outflow (Φ_{out}) via Tachyonic Projection:** To prevent thermal runaway and over-pressurization from the continuous multidimensional inflow, the D_3 vacuum utilizes its global cosmological horizon as a drainage valve. Because our D_3 manifold resides within a higher-dimensional black hole interior, the thermodynamic outflow Φ_{out} does not manifest internally as standard Hawking radiation. Instead, overloaded informational states undergo a critical phase separation at the boundary, where excess energy is deconstructed and projected out of the active 3D metric sheet into the surrounding D_4 bulk. This outward mass-energy leak manifests across the cosmological boundary as a superluminal, tachyonic loss gradient, maintaining the global thermodynamic homeostasis of our internal universe by shedding excess entropic stress into the higher-dimensional environment.

- **Origin of the Cosmic Microwave Background (CMB) without Inflation:** The near-perfect isotropy of the Cosmic Microwave Background ($\Delta T/T \approx 10^{-5}$) is traditionally explained by cosmic inflation. Within our multi-sheeted framework, we propose a direct physical alternative: the CMB is the thermal signature of the continuous, isotropic mass-energy influx Φ_{in} entering our D_3 manifold through the global, spherical S^2 boundary layer of the D_4 event horizon. Because the $D_4 \rightarrow D_3$ mass transfer is regulated by the uniform, holographic surface gravity of the parent D_4 star, the injected energy is thermalized homogeneously across the entire boundary before propagating inward. This constant, global thermodynamic feed naturally establishes a uniform thermal baseline across all causal horizons, resolving the classical cosmological horizon problem without requiring an inflationary epoch.

8.2. The Hubble Tension Quantitative Resolution

The Hubble tension is resolved by demonstrating that the temporal coordinate $\tau(x)$ scales non-linearly due to informational equivalence: $\tau(x) = \tau_0 \cdot x^{-1/2}$ [4], where τ_0 defines the fundamental baseline temporal scale of the coordinate metric. Incorporating Jacobson’s thermodynamic approach ($\delta Q = TdS$), the local metric of the numerical vacuum curves under the pressure of the dimensional transition [4]:

$$ds^2 = \tau(x)^2 dx^2 = \frac{\tau_0^2}{x} dx^2 \quad (17)$$

Integrating this curved metric yields the true, information-scaled physical distance $S(x) = 2\tau_0 \sqrt{x}$ [4]. The cosmological redshift is therefore not a pure Doppler effect of recessional velocity, but an accumulated phase shift within the curved, self-reactive number space [7].

The total scaling domain of the D_3 manifold is bounded by the initialization threshold of the Radix-6 engine ($x_{\text{init}} = 6$) and the critical collapse boundary of the active Radix-210 state machine ($x_{\text{crit}} = 211$, corresponding to the first prime number outside the stable Π_4 system) [7]:

$$\chi = \frac{x_{\text{crit}}}{x_{\text{init}}} = \frac{211}{6} \approx 35.1667 \quad (18)$$

The global metric deformation factor Δ_{metric} represents the logarithmic expansion gradient scaled by the transcendent phase-damping factor $2\pi \cdot e$ of the D_2 complex plane [7]:

$$\Delta_{\text{metric}} = \frac{\ln(\chi)}{2\pi \cdot e} \approx 0.20843 \quad (19)$$

When mapping global, early-universe coordinates onto local, late-universe observers via the stellar distance modulus (corrected by $\ln(10)$), the apparent expansion rate scales according to [6, 8]:

$$H_0^{\text{Calculated}} = H_0^{\text{Planck}} \cdot \left(1 + \frac{\Delta_{\text{metric}}}{\ln(10)}\right) \quad (20)$$

Substituting the empirical value of $H_0^{\text{Planck}} = 67.4$ km/s/Mpc yields $H_0^{\text{Calculated}} \approx 73.50$ km/s/Mpc, falling precisely within the 1σ error margin of the local SH0ES measurements (73.04 ± 1.04 km/s/Mpc) [8].



Fig. 4: A diving analogy illustrating Hubble redshift and matter density. **Left:** High matter density (underwater), where the medium heavily absorbs and filters light. **Center Split (M1/M2):** The transition phase (underwater vs. surface boundary) illustrating varying density gradients. **Right:** The zero-density limit (beach/above water) representing the Hubble constant in a medium-free, uninhibited expansion limit.

8.3. Dimensional Scaling of Fundamental Constants

The embedding of our 3D observable universe as a holographic membrane on the S^3 event horizon of a collapsing 4D star requires a systematic re-evaluation of fundamental constants [9]:

Instead of representing gravity as an absolute invariant coupling, we define the effective three-dimensional gravitational parameter $G_{3D}(t)$ as a dynamic state variable modulated by the volumetric dilution of the elastic vacuum medium. Let κ_0 be the fundamental structural tension modulus of the unperturbed D_4 carrier lattice. The emergent gravitational coupling depends on the background vacuum density $\rho_{\text{vacuum}}(t)$ and the cosmic scale factor $a(t)$ according to:

$$G_{3D}(t) = \frac{G_0}{1 + \gamma_{\text{relax}} \left(\frac{\rho_{\text{crit}}}{\rho_{\text{vacuum}}(t)} \right)} \equiv G_0 \cdot [1 - \beta_{\text{elas}} \ln(a(t))] \quad (21)$$

where G_0 represents the primordial coupling limit at the boundary of the topological phase transition, while γ_{relax} and β_{elas} are non-dimensional scaling constants reflecting the progressive elastic relaxation of the superfluid matrix. As the D_3 manifold expands ($a(t) > 1$), the mechanical counter-pressure drops, naturally revealing the macroscopically observed weakness of gravity as an entropic aging signature of the spatial lattice [9].

- **Planck’s Constant as a Phase Boundary ($\hbar$):** Represents the fundamental phase-action limit of the microscopic poloidal coordinate r , restricted by the minimum loop path of the toroidal vacuum sheet [10].
- **Resolution of the Cosmological Constant Problem (Λ):** We resolve the greatest discrepancy in modern physics (120 orders of magnitude) [9]. Λ is not an intrinsic zero-point energy density of flat space, but the physical density of the ongoing 4D mass influx $\dot{M}_{4D}$ distributed across the S^3 boundary volume $V_{3D} = 2\pi^2 R^3$ [9]:

$$\Lambda(t) \equiv 8\pi G_{3D}(t) \rho_{\text{vacuum}}(t) \quad (22)$$

where $\rho_{\text{vacuum}}(t)$ describes the time-dependent local coordinate density of the vacuum medium. This links Λ directly to the active, finite thermodynamic mass-energy exchange between adjacent dimensional sheets [9].

9. Discussion and Conclusion

The synthesis of the sequential dimensional cascade from D_0 to D_4 establishes a self-consistent, strictly non-singular paradigm that shifts cosmology from passive geometric modeling to active vacuum thermodynamics. By mapping the fundamental physical properties of spin ($p = 2$), structural form ($p = 3$), and mass ($p = 5$) onto prime numbers, physical phenomena are revealed to be emergent resonance states of the underlying Radix-210 state machine.

A critical refinement within this framework concerns the dynamic behavior of the fine-structure constant $\alpha(t)$. Rather than driving an asymptotic cosmic collapse through pure volume expansion, $\alpha(t)$ serves as a direct indicator of the vacuum’s local phase compression and elastic tension. In the unperturbed, highly diluted limit of maximum spatial expansion, the vacuum relaxation dictates $\alpha \rightarrow 0$, signifying the decoupling of localized electromagnetic fields. Conversely, the critical boundary $\alpha \rightarrow 1$ represents the absolute saturation limit of spatial compaction, achieved exclusively at the core of nucleonic bonds ($p = 5$) and macroscopic singularities.

Consequently, the lifetime of the universe is not bounded by an arbitrary temporal clock, but by the thermodynamic stability of the $x = 6$ Stability Wave (2×3) along the critical Riemann boundary $\text{Re}(s) = 1/2$. The cosmos exists in a homeostatically regulated, global lifecycle:

1. **Genesis and Influx:** Mass-energy enters the D_3 manifold via isotropic influx (Φ_{in}) at the global horizon boundary, mapping out the uniform baseline of the Cosmic Microwave Background without requiring an inflationary mechanism.
2. **Crystallization and Balance:** This influx scales through the primordial radix engines, materializing into bulk baryonic mass ($p = 5$) that is prevented from immediate gravitational implosion by the structural rigidity and outward geometric pressure of the $x = 6$ standing resonance wave.
3. **Entropic Drainage and Decay:** As localized regions accumulate information density beyond the Harmonic Saturation Limit of $\phi(210) = 48$, the local phase state is forced to deviate from the $\text{Re}(s) = 1/2$ Riemann equilibrium line. This induces a localized collapse of the $x = 6$ rigid network, triggering the uncompensated thermodynamic implosion of $p = 5$ mass profiles.

700 Through this mechanism, macroscopic event horizons (black holes) and potential localized false vacuum decays do not represent absolute endpoints of existence, but rather crucial thermodynamic drainage valves (Φ_{out}). They losslessly deconstruct high-entropy matter back into its dimensionless ($0D$) information potential, closing the cosmic circle by re-feeding the primordial D_0 phase space.

710 In conclusion, this framework successfully resolves the Hubble tension as a natural projection artifact of a density-modulated, self-reactive number space, while simultaneously deriving the theoretical limits of light speed and quantum orbital capacities. The Radix-210 vacuum model proves that the laws of physics and the distribution of matter are structural inevitabilities of analytic number theory, orchestrating a homeostatically balanced, eternal thermodynamic lifecycle.

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